Exercises: Kinematics

IBDP Physics SL · Grade 11 · A.1 Kinematics · 24 calculation exercises in three IB levels

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Name: Class: Date:

How to use. Show your working: equation → substitution with units → answer with unit and sensible significant figures. Use g = 9.81 m s^{-2}. Level 1 = one step (Paper 1 style), Level 2 = multi-step, Level 3 = structured exam questions with marks.
Check yourself online (instant marking, hints and worked solutions, new numbers to practise again):
https://school.taneski.com/physics/dp/ex-kinematics/

Level 1 · Paper 1 style — one step

8 exercises · 16 marks · show your working

1.1 Accelerating car

A.1 · Equations of motion · [2 marks]

A car accelerates uniformly from rest to 26 m s^{-1} in 9.0 s. Calculate its acceleration.

Answer: _____ m s^{-2}

1.2 Cycling to school

A.1 · Equations of motion · [2 marks]

A cyclist rides at a constant 20 km h^{-1}. Calculate the time to ride 2.8 km. Give your answer in seconds.

Answer: _____ s

1.3 Dropped stone

A.1 · Free fall · [2 marks]

A stone is dropped from rest from a bridge 30 m above a river. Air resistance is negligible. Calculate the time it takes to reach the water.

Answer: _____ s

1.4 Impact speed

A.1 · Free fall · [2 marks]

A ball is dropped from rest from a height of 3.6 m. Air resistance is negligible. Calculate its speed just before it hits the ground.

Answer: _____ m s^{-1}

1.5 Braking train

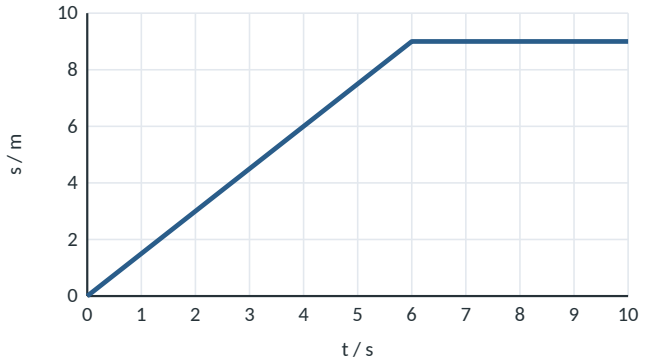
A.1 · Equations of motion · [2 marks]

A train travelling at 32 m s^{-1} decelerates uniformly at 1.7 m s^{-2} until it stops. Calculate the stopping distance.

Answer: _____ m

1.6 Gradient of a displacement–time graph A.1 · Motion graphs · [2 marks]

The graph shows the displacement of a walker against time. Calculate the velocity of the walker during the first 6.0 s.



Answer: _____ m s^{-1}

1.7 Velocity components A.1 · Projectiles · [2 marks]

A ball is kicked at 20 m s^{-1} at 45° above the horizontal.

(a) Calculate the horizontal component of the initial velocity. [1]

(a) Answer: _____ m s^{-1}

(b) Calculate the vertical component of the initial velocity. [1]

(b) Answer: _____ m s^{-1}

1.8 Average speed of a lap runner A.1 · Equations of motion · [2 marks]

A runner completes 4 laps of a 400 m track in 160 s and finishes where she started.

(a) Calculate her average speed. [1]

(a) Answer: _____ m s^{-1}

(b) State her average velocity. Explain your answer. [1]

Level 2 · Multi-step — two or three linked steps

8 exercises · 33 marks · show your working

2.1 Ball thrown upwards A.1 · Free fall · [4 marks]

A ball is thrown vertically upwards at 17 m s^{-1} and caught again at the same height. Air resistance is negligible.

(a) Calculate the maximum height reached. [2]

(a) Answer: _____ m

(b) Calculate the total time the ball is in the air. [2]

(b) Answer: _____ s

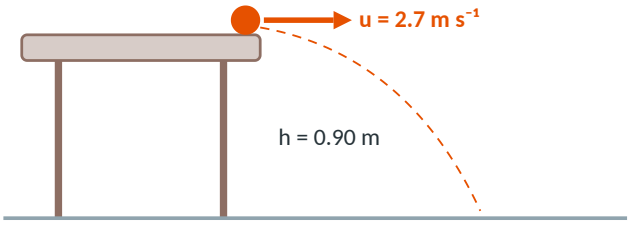
2.2 Stopping distance A.1 · Equations of motion · [4 marks]

A driver travelling at 27 m s^{-1} sees a hazard. Her reaction time is 0.80 s, then the brakes decelerate the car uniformly at 7.0 m s^{-2}. Calculate the total stopping distance.

Answer: _____ m

2.3 Rolling off a table A.1 · Projectiles · [5 marks]

A ball rolls off a horizontal table 0.90 m high at 2.7 m s^{-1}. Air resistance is negligible.



(a) Calculate the time the ball is in the air. [2]

(a) Answer: _____ s

(b) Calculate the horizontal distance from the table edge to where the ball lands. [1]

(b) Answer: _____ m

(c) Calculate the speed of the ball just before it lands. [2]

(c) Answer: _____ m s^{-1}

2.4 Stone thrown down from a bridge A.1 · Free fall · [4 marks]

A stone is thrown vertically downwards at 6.0 m s^{-1} from a bridge 22 m above the water. Air resistance is negligible.

(a) Calculate the speed of the stone as it reaches the water. [2]

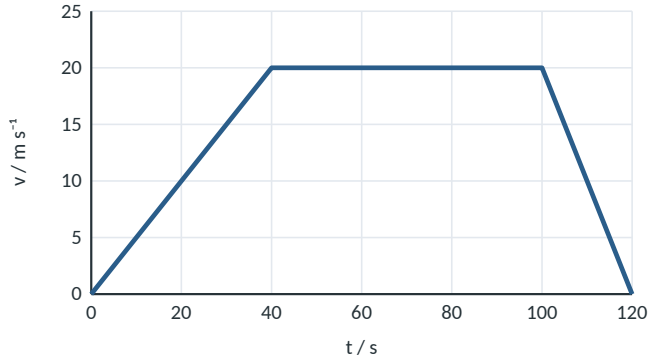
(a) Answer: _____ m s^{-1}

(b) Calculate the time taken to reach the water. [2]

(b) Answer: _____ s

2.5 Train journey from a v–t graph A.1 · Motion graphs · [4 marks]

The graph shows the velocity of a train between two stations.



(a) Calculate the acceleration during the first 40 s. [1]

(a) Answer: _____ m s^{-2}

(b) Calculate the distance between the stations. [2]

(b) Answer: _____ m

(c) Calculate the average speed for the whole journey. [1]

(c) Answer: _____ m s^{-1}

2.6 Kicked ball on level ground A.1 · Projectiles · [5 marks]

A football is kicked from level ground at 22 m s^{-1} at 40° above the horizontal. Air resistance is negligible.

(a) Calculate the time of flight. [2]

(a) Answer: _____ s

(b) Calculate the horizontal range. [2]

(b) Answer: _____ m

(c) Calculate the maximum height. [1]

(c) Answer: _____ m

2.7 Stone thrown from a cliff A.1 · Projectiles · [4 marks]

A stone is thrown horizontally at 14 m s^{-1} from the top of a cliff 40 m above the sea. Air resistance is negligible.

(a) Calculate the horizontal distance from the foot of the cliff to where the stone lands. [2]

(a) Answer: _____ m

(b) Calculate the angle below the horizontal at which the stone hits the sea. [2]

(b) Answer: _____ °

2.8 Measuring g with a light gate A.1 · Free fall · [3 marks]

In a lab a ball is released from rest 1.25 m above a light gate. The timer records a fall time of 0.499 s.

(a) Calculate the value of g from these data. [2]

(a) Answer: _____ m s^{-2}

(b) Calculate the percentage difference from 9.81 m s^{-2}. [1]

(b) Answer: _____ %

Level 3 · Paper 2 structured — parts and marks

8 exercises · 58 marks · show your working

3.1 Police chase A.1 · Equations of motion · [8 marks]

A speeding car passes a parked police car at a constant 16 m s^{-1}. 2.0 s later the police car sets off from rest with a constant acceleration of 3.2 m s^{-2}. Take t = 0 when the speeding car passes the police car.

(a) Calculate how far ahead the speeding car is when the police car sets off. [1]

(a) Answer: _____ m

(b) Show that the police car catches the speeding car about 14 s after the speeder passed it. [3]

(b) Answer: _____ s

(c) Calculate the distance from the start at which this happens. [1]

(c) Answer: _____ m

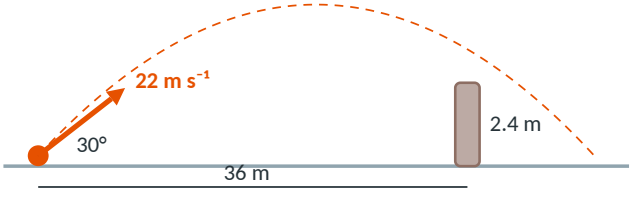
(d) Calculate the speed of the police car at that moment. [1]

(d) Answer: _____ m s^{-1}

(e) Sketch, on the same axes, the velocity–time graphs of both cars and state how the graphs show that the two cars have the same displacement when the police car catches up. [2]

3.2 Clearing a wall A.1 · Projectiles · [7 marks]

A football is kicked from the ground at 22 m s^{-1} at 30° above the horizontal towards a wall 36 m away. The wall is 2.4 m high. Air resistance is negligible.



(a) Calculate the horizontal component of the initial velocity. [1]

(a) Answer: _____ m s^{-1}

(b) Calculate the time taken for the ball to reach the wall. [1]

(b) Answer: _____ s

(c) Calculate the height of the ball as it reaches the wall. [3]

(c) Answer: _____ m

(d) State whether the ball clears the wall, and whether it is rising or falling as it passes the wall. Justify both. [2]

3.3 Model rocket A.1 · Free fall · [9 marks]

A model rocket is launched vertically from rest. Its motor gives it a constant upward acceleration of 12 m s^{-2} for 3.0 s, then the fuel runs out. After that the rocket moves freely under gravity. Air resistance is negligible.

(a) Calculate the speed of the rocket when the fuel runs out. [1]

(a) Answer: _____ m s^{-1}

(b) Calculate the height of the rocket when the fuel runs out. [1]

(b) Answer: _____ m

(c) Calculate the maximum height reached by the rocket. [2]

(c) Answer: _____ m

(d) Calculate the total time from launch until the rocket hits the ground. [3]

(d) Answer: _____ s

(e) Sketch the velocity–time graph for the whole flight (up positive). Label the gradient of each stage. [2]

3.4 Measuring g — uncertainties A.1 · Free fall · [7 marks]

A student drops a ball from rest through a measured height h = 1.50 ± 0.01 m and times the fall with light gates: t = 0.555 ± 0.005 s.

(a) Calculate g from these data. [2]

(a) Answer: _____ m s^{-2}

(b) Calculate the fractional uncertainty in g. [2]

(b) Answer: _____

(c) Calculate the absolute uncertainty in g. [1]

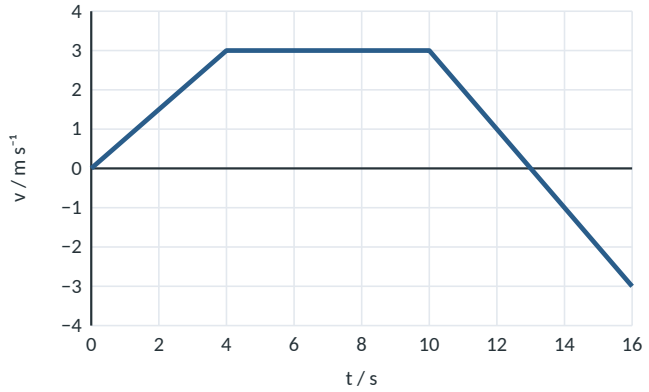
(c) Answer: _____ m s^{-2}

(d) State whether the result agrees with 9.81 m s^{-2}, and suggest which measurement should be improved first. Explain. [2]

3.5 Lift journey from a v–t graph

A.1 · Motion graphs · [6 marks]

The graph shows the velocity of a lift. Upwards is positive.



(a) Calculate the acceleration of the lift between t = 0 and t = 4.0 s. [1]

(a) Answer: _____ m s^{-2}

(b) Calculate the acceleration between t = 10 s and t = 16 s. [1]

(b) Answer: _____ m s^{-2}

(c) Calculate the displacement of the lift after 16 s. [2]

(c) Answer: _____ m

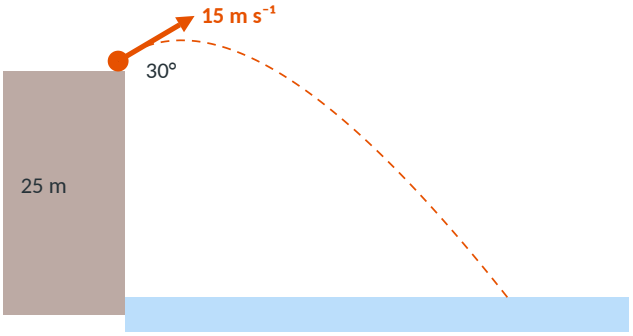
(d) Calculate the total distance travelled in the 16 s. [1]

(d) Answer: _____ m

(e) Describe the motion of the lift between t = 13 s and t = 16 s. [1]

3.6 Throw from a cliff top A.1 · Projectiles · [7 marks]

A ball is thrown from the top of a cliff 25 m above the sea at 15 m s^{-1}, 30° above the horizontal. Air resistance is negligible.



(a) Calculate the vertical and horizontal components of the initial velocity. Enter the vertical one. [1]

(a) Answer: _____ m s^{-1}

(b) Show that the ball is in the air for about 3.2 s. [3]

(b) Answer: _____ s

(c) Calculate the horizontal distance travelled by the ball. [1]

(c) Answer: _____ m

(d) Calculate the speed of the ball as it hits the sea. [2]

(d) Answer: _____ m s^{-1}

3.7 Thrown up on the Moon A.1 · Free fall · [7 marks]

An astronaut throws a tool vertically upwards at 6.0 m s^{-1} on the Moon, where g_{Moon} = 1.62 m s^{-2}. On Earth the same throw would use g = 9.81 m s^{-2}.

(a) Calculate the maximum height reached on the Moon. [2]

(a) Answer: _____ m

(b) Calculate the time the tool takes to return to the astronaut's hand on the Moon. [2]

(b) Answer: _____ s

(c) Calculate how many times higher the tool goes on the Moon than on Earth. [2]

(c) Answer: _____

(d) Explain why air resistance does not need to be considered on the Moon. [1]

3.8 Braking-distance data

A.1 · Motion graphs · [7 marks]

A road-safety test measures the braking distance s of a car for different initial speeds u (same road, same deceleration):

u / m s^{-1}	10	15	20	25	30
s / m	7.7	17.3	30.8	48.1	69.2

(a) Show that s is proportional to u^2 by explaining which graph you would plot. [2]

(b) Determine the gradient of the graph of s against u^2 (use the first and last data points). [2]

(b) Answer: _____ s^2 m^{-1}

(c) Determine the deceleration of the car. [2]

(c) Answer: _____ m s^{-2}

(d) Predict the braking distance at 35 m s^{-1}. [1]

(d) Answer: _____ m

End of exercises